Set—**01**

**Shahjalal University of Science and Technology**
Department of Computer Science and Engineering
$3^{rd}$ year $2^{nd}$ Semester Final Examination—June 2020 (Session 2017-18)
Course No.—**CSE 329**
Course Title—**Computer Architecture**
Time—**5 Hours** Credit: **3.00** Total Marks#**30**

(Answer All the Questions)

## Group A

1. Answer the following **Five** Questions in short. $5 \times 1 = 5$
   (a) What is the advantage of using virtual memory?
   (b) According to the trend in device scaling historically observed by Moore's Law, the number of transistors on a chip in 2025 should be how many times the number in 2020?
   (c) What is the stored programming concept?
   (d) What MIPS instruction does this represent?

| **op** | **rs** | **rt** | **rd** | **shamnt** | **funct** |
|---|---|---|---|---|---|
| 0 | 18 | 8 | 8 | 0 | 32 |

   (e) State Amdahl's Law

2. Answer the following **Two** Questions. $2 \times 2.5 = 5$
   (a) Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for some program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program. Which computer is faster for this program and by how much?
   (b) What is the difference between RISC and CISC? Mention the key advantages and disadvantages of RISC and CISC.

3. Design a 1-bit ALU that can perform (A AND B). (A OR B) and A+B. Using 1-bit ALU design a 32-bit ALU, where A and B are 32-bit numbers. 5

## Group B

1. Answer the following **Five** Questions. $5 \times 1 = 5$
   (a) What do you mean by ISA?
   (b) What do you mean by opcode?
   (c) Negate $3_{ten}$ and then check the result by negating $-3_{ten}$. (use 8 bits)
   (d) What do you mean by Computer Architecture?
   (e) If computer A runs a program in 10 seconds and computer B runs the same program in 15 seconds, how much faster is A than B?

2. Answer the following **Two** Questions. $2 \times 2.5 = 5$
   (a) In the following code segment, f, g, h, i, and j are variables. If the five variables f through j correspond to the five registers $s0 through $s4, what is the compiled MIPS code for this C if statement?

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if (i == j) f = g + h; else f = g - h;
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   (b) A program runs in 10 seconds on computer A, which has a 2 GHz clock. We are trying to help a computer designer build a computer, B, which will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target?

3. Show the IEEE 754 binary representation of the number $-0.75_{ten}$ in single and double precision. 5

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